

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. Branch : Electronics and Computer Engineering Semester : III

Subject Code & Name: BTESC305 & Digital Electronics and Microprocessor

Max Marks: 60M

Date: 17/03/2023

Duration: 03:00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12M
A) Perform following Conversion. a. $(11100111)_2 = (?)_{\text{gray}}$ b. $(1792.12)_{10} = (?)_2$ c. $(338.025)_{10} = (?)_8$	BL2 & CO1	6
B) Explain the working of following gates with their truth table and logic diagram. a) AND b) NOR c) EX-OR	BL2 & CO1	6
C) Write a short note on multiplexer and demultiplexer.	BL2 & CO1	6
Q.2 Solve Any Two of the following.		12M
A) Minimize the following equation using K-map. a. $Y = \sum m(0,2,4,7,8,9,11,12,13)$ b. $Y = \prod M(0,1,2,3,5,7)$	BL3 & CO2	6
B) Explain the working of full subtractor with truth table. Implement it with half subtractor.	BL2 & CO2	6
C) Design 2 bit digital comparator using logic gates.	BL3 & CO2	6
Q. 3 Solve Any Two of the following.		12M
A) Explain S-R flip flop with logic diagram and truth table.	BL3 & CO3	6
B) What is counter? Explain 4-bit ring counter using D-flip flop. Implement its truth table.	BL2 & CO3	6
C) Write a short note on shift register and list down its application.	BL3 & CO3	6
Q.4 Solve Any Two of the following.		12M
A) Draw and explain block diagram of 8086 microprocessor and explain in detail.	BL1 & CO4	6
B) Explain the addressing modes of 8086 with suitable example.	BL2 & CO4	6
C) Explain the structure of 8086 PSW	BL3 & CO4	6

Q. 5 Solve Any Two of the following.

12M

A) Explain with example arithmetic group of instruction set in 8085.

BL3& CO5

6

B) Compare 8085 with 8086.

BL3& CO5

6

C) Write a program addition of two 16 bit number using 8085

BL3& CO5

6

***** End *****